

CMPE 321 – Project 1: TransferDB

Part 1: Conceptual Database Design

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1 Introduction

TransferDB is a database system designed to manage a football ecosystem, covering players, managers, referees, clubs, stadiums, contracts, transfers, competitions, and matches. This report presents the conceptual design of TransferDB using the Entity-Relationship (ER) model, identifying all entity sets, relationship sets, key constraints, participation constraints, and ISA hierarchies.

2 Entity Sets

2.1 Person

Supertype entity representing all individuals in the system.

- **person_id** (Primary Key)
- name
- surname
- nationality
- date_of_birth

ISA Hierarchy: Person has three subtypes: Player, Manager, and Referee.

- **Disjoint:** A person can hold only one role at a time.
- **Covering:** Every person must be exactly one of Player, Manager, or Referee.

2.2 Player (ISA Person)

- market_value (in Euros, must be > 0)

- `main_position` (Goalkeeper, Defender, Midfielder, or Forward)
- `strong_foot` (Right, Left, or Both)
- `height` (in cm, positive integer)

2.3 Manager (ISA Person)

- `preferred_formation` (e.g., 4-3-3, 4-4-2)
- `experience_level` (e.g., Beginner, Intermediate, Elite)

2.4 Referee (ISA Person)

- `license_level` (e.g., FIFA, Continental, National)
- `years_of_experience` (non-negative integer)

2.5 Club

- `club_id` (Primary Key)
- `club_name` (Unique)
- `city`
- `foundation_year`

2.6 Stadium

- `stadium_id` (Primary Key)
- `stadium_name`
- `city`
- `capacity` (positive integer)

Uniqueness constraint: The combination of (`stadium_name`, `city`) must be unique.

2.7 Contract

A formal agreement binding a Player to a Club. History is preserved; records are never deleted.

- `contract_id` (Primary Key)
- `start_date`
- `end_date` (must be strictly later than `start_date`)
- `weekly_wage` (in Euros, must be > 0)
- `contract_type` (Permanent or Loan)

2.8 Transfer Record

Permanently stored history of all player movements between clubs.

- **transfer_id** (Primary Key)
- transfer_date
- transfer_fee (in Euros, must be ≥ 0)
- transfer_type (Free, Purchase, or Loan)

2.9 Competition

- **competition_id** (Primary Key)
- name
- season (e.g., 2024/2025)
- country
- competition_type (e.g., League, Cup, International)

Uniqueness constraint: The combination of (name, season) must be unique.

2.10 Match

- **match_id** (Primary Key)
- match_datetime
- attendance (non-negative integer)
- home_goals (non-negative integer)
- away_goals (non-negative integer)

The match result (Home Win, Away Win, Draw) is derived from the scores and is not stored.

3 Relationship Sets

In the notation below:

- **Key constraint** = an arrow in the ER diagram (at most one participation).
- **Total participation** = a bold/thick line (every entity must participate).

3.1 leads (Manager – Club)

- Manager side: key constraint (a manager manages at most one club)
- Club side: key constraint + total participation (every club must have exactly one manager)
- This is a 1-to-1 relationship with total participation on the Club side.

3.2 primary_venue (Club – Stadium)

- Club side: key constraint + total participation (every club has exactly one home stadium)
- Stadium side: no constraint (multiple clubs may share a stadium)
- Many-to-1 from Club to Stadium.

3.3 signs (Player – Contract)

- Contract side: key constraint + total participation (every contract belongs to exactly one player)
- Player side: no constraint (a player may have many contracts over a career)

3.4 with (Contract – Club)

- Contract side: key constraint + total participation (every contract is with exactly one club)
- Club side: no constraint

3.5 including (Player – Transfer Record)

- Transfer Record side: key constraint + total participation (every transfer involves exactly one player)
- Player side: no constraint

3.6 from (Transfer Record – Club)

- Transfer Record side: key constraint + total participation (every transfer has a source club)
- Club side: no constraint
- The from_club and to_club must be different clubs.

3.7 to (Transfer Record – Club)

- Transfer Record side: key constraint + total participation (every transfer has a destination club)
- Club side: no constraint

3.8 participates (Club – Competition)

- Many-to-many relationship with no key constraints.
- A club may participate in multiple competitions, and a competition has multiple clubs.

3.9 belongs_to (Match – Competition)

- Match side: key constraint + total participation (every match belongs to exactly one competition)
- Competition side: no constraint

3.10 home (Match – Club)

- Match side: key constraint + total participation (every match has exactly one home club)
- Club side: no constraint

3.11 away (Match – Club)

- Match side: key constraint + total participation (every match has exactly one away club)
- Club side: no constraint
- The home and away clubs must be different.

3.12 played_at (Match – Stadium)

- Match side: key constraint + total participation (every match is played at exactly one stadium)
- Stadium side: no constraint
- The stadium does not need to be the home club's primary venue.

3.13 officiates (Referee – Match)

- Match side: key constraint + total participation (every match has exactly one referee)
- Referee side: no constraint

3.14 match_stats (Player – Match)

- Many-to-many relationship with total participation on the Match side.

- Descriptive attributes: is_starter, minutes_played (0–120), position_in_match, goals (≥ 0), assists (≥ 0), yellow_cards (0–2), red_cards (Boolean), rating (1.0–10.0).

4 ER Diagram

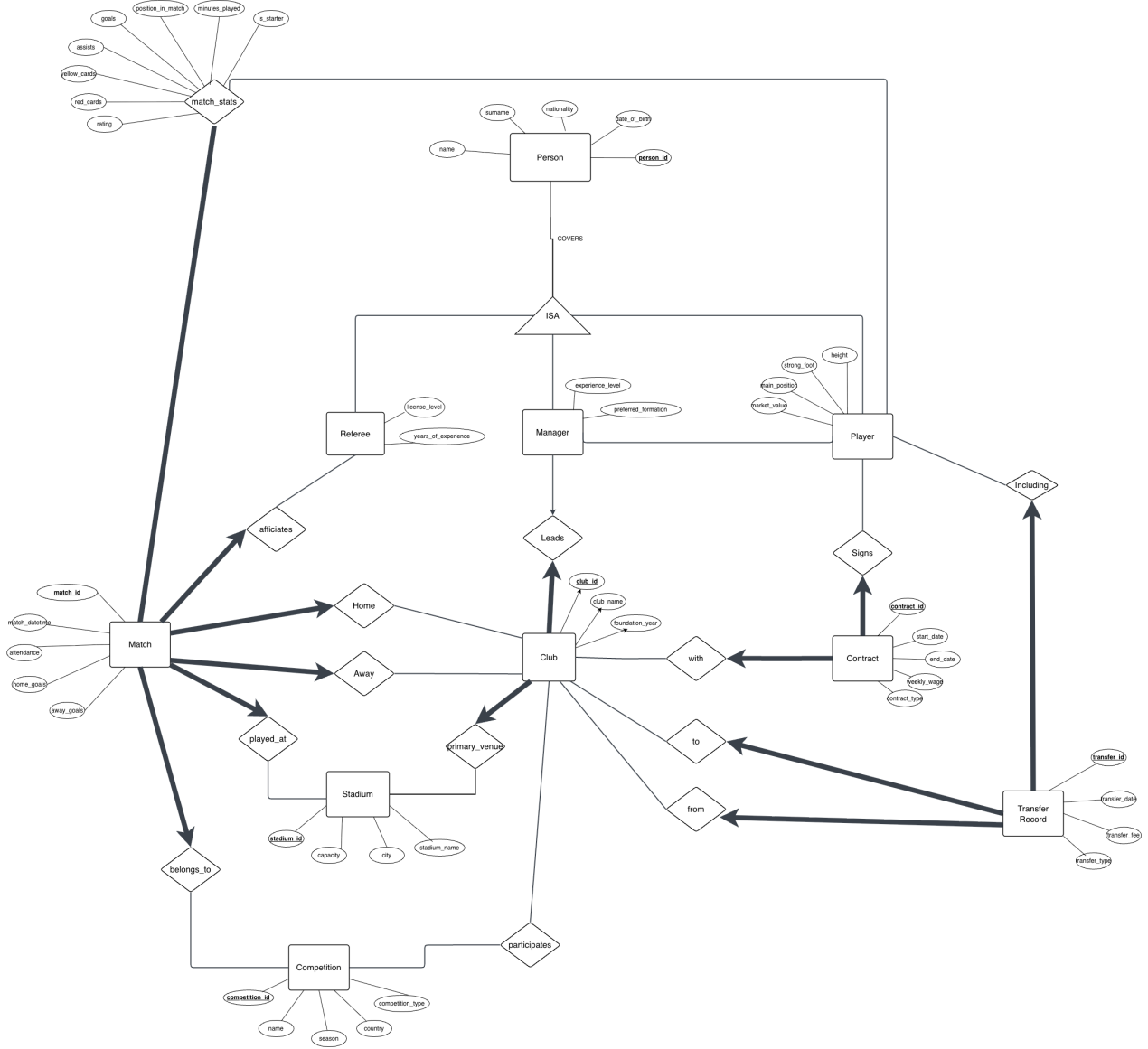


Figure 1: ER Diagram for TransferDB

5 Discussion: Constraints Not Represented in the ER Diagram

The following constraints from the project description cannot be fully captured using the ER model notation and will need to be enforced at the application level or documented as known limitations.

5.1 Contract Constraints

- A player’s current club is derived from their active contracts (where $\text{start_date} \leq \text{current date} \leq \text{end_date}$). The concept of an “active” contract is temporal and cannot be represented structurally.
- A player who has no active contract is considered a free agent. This is a derived state.
- A player must not have two active contracts of the same type running concurrently. This requires checking overlapping date ranges across multiple contract records.
- A Loan contract is only valid if the player holds an active Permanent contract with a *different* club at the same time. This requires cross-entity temporal validation.

5.2 Transfer Record Constraints

- For a Loan transfer, the `from_club` must match the player’s current parent club (the club with the active Permanent contract). This is a cross-entity constraint.
- A Loan transfer requires the player to have an active Permanent contract. This involves temporal checks.
- For a Free transfer, the transfer fee must be exactly 0. For Purchase and Loan transfers, the fee must be greater than 0. While this is a domain constraint on Transfer Record, the ER model does not express conditional value restrictions.

5.3 Club Constraints

- The squad of a club (the set of players currently under contract) is derived from active contracts, not directly modeled as a stored relationship.

5.4 Match Scheduling Constraints

- A match has a fixed duration of 120 minutes. No two matches may be played at the same stadium if their start times are within 120 minutes of each other.

- A club may not appear in two matches (as either home or away) whose start times are within 120 minutes of each other.
- A referee may not be assigned to two matches whose start times are within 120 minutes of each other.
- These are temporal cross-entity constraints that the ER model cannot express.

5.5 Attendance Constraint

- The attendance of a match must not exceed the capacity of the stadium where the match is played. This is a cross-entity constraint between Match and Stadium.

5.6 Match Result

- The outcome of a match (Home Win, Away Win, Draw) is derived by comparing home_goals and away_goals. It is not stored as a separate attribute, by design.

5.7 Match Participation and Squad Rules

- Total participation of Match in the match_stats relationship (every match must have at least one player participating) cannot be enforced structurally in many-to-many relationships.
- A maximum of 11 players may start a match for the same club. This constraint requires counting participating entities and therefore cannot be represented using standard ER notation.
- Each club may register at most 23 players for a match squad. This constraint requires counting the number of participating players for a club in a match and cannot be represented directly in the ER diagram. constraint and cannot be expressed in the ER diagram.
- A player cannot appear in a match as both a home team and an away team participant. This requires cross-row validation.
- A player who receives 2 yellow cards in a single match receives a red card automatically. This is a derived/triggered constraint.
- A player who receives a red card is automatically suspended for the club's next match in the same competition. This requires querying historical match data.
- A player who accumulates 5 yellow cards across a single season in a given competition is suspended for the club's next match. This requires aggregation across multiple matches.

5.8 Domain Value Constraints

The ER model cannot express attribute value ranges, allowed value sets, or conditional value restrictions. The following constraints must be enforced at the logical level:

- `market_value > 0`, `height > 0` (Player)
- `capacity > 0` (Stadium)
- `years_of_experience ≥ 0` (Referee)
- `weekly_wage > 0`, `end_date > start_date` (Contract)
- `transfer_fee ≥ 0` (Transfer Record)
- `attendance ≥ 0`, `home_goals ≥ 0`, `away_goals ≥ 0` (Match)
- `minutes_played ∈ [0, 120]`, `goals ≥ 0`, `assists ≥ 0`, `yellow_cards ∈ [0, 2]`, `rating ∈ [1.0, 10.0]` (Match Stats)
- `main_position` must be one of: Goalkeeper, Defender, Midfielder, Forward
- `strong_foot` must be one of: Right, Left, Both
- `contract_type` must be one of: Permanent, Loan
- `transfer_type` must be one of: Free, Purchase, Loan
- `home_club` and `away_club` must refer to different clubs
- `from_club` and `to_club` must refer to different clubs

5.9 Uniqueness Constraints

The following uniqueness constraints go beyond primary keys and cannot be fully expressed in ER notation:

- `club_name` must be unique across all clubs.
- The combination of (`stadium_name`, `city`) must be unique.
- The combination of (competition name, season) must be unique.